

Unit 7: The Derivative at a Point — Full Solutions

Warm-up

1. $f(x) = x^2 + 5x$, find $f'(1)$.

$$f(1) = 1 + 5 = 6.$$

$$f(1+h) = (1+h)^2 + 5(1+h) = 1 + 2h + h^2 + 5 + 5h = 6 + 7h + h^2.$$

Subtract $f(1) = 6$: $7h + h^2$.

Divide by h : $7 + h$.

Let $h \rightarrow 0$: 7.

$$f'(1) = 7.$$

2. $f(x) = 3x^2 - 2$, find $f'(0)$.

$$f(0) = -2.$$

$$f(0+h) = 3h^2 - 2.$$

Subtract $f(0) = -2$: $3h^2$.

Divide by h : $3h$.

Let $h \rightarrow 0$: 0.

$$f'(0) = 0.$$

3. $f(x) = -x^2 + 4x$, find $f'(2)$.

$$f(2) = -4 + 8 = 4.$$

$$f(2+h) = -(2+h)^2 + 4(2+h) = -(4 + 4h + h^2) + 8 + 4h = -4 - 4h - h^2 + 8 + 4h = 4 - h^2.$$

Subtract $f(2) = 4$: $-h^2$.

Divide by h : $-h$.

Let $h \rightarrow 0$: 0.

$f'(2) = 0$. (This makes sense: $-x^2 + 4x$ is a downward parabola with its peak exactly at $x = 2$, and the tangent line at the very top of a hill is always flat.)

4. $f(x) = 5x - x^2$, find $f'(-1)$.

$$f(-1) = -5 - 1 = -6.$$

$$f(-1+h) = 5(-1+h) - (-1+h)^2 = -5 + 5h - (1 - 2h + h^2) = -5 + 5h - 1 + 2h - h^2 = -6 + 7h - h^2.$$

Subtract $f(-1) = -6$: $7h - h^2$.

Divide by h : $7 - h$.

Let $h \rightarrow 0$: 7.

$$f'(-1) = 7.$$

5. $f(x) = 2x^2 + 3$, find $f'(3)$.

$$f(3) = 18 + 3 = 21.$$

$$f(3+h) = 2(3+h)^2 + 3 = 2(9+6h+h^2) + 3 = 18 + 12h + 2h^2 + 3 = 21 + 12h + 2h^2.$$

$$\text{Subtract } f(3) = 21: 12h + 2h^2.$$

$$\text{Divide by } h: 12 + 2h.$$

$$\text{Let } h \rightarrow 0: 12.$$

$$f'(3) = 12.$$

$$6. f(x) = x^2 - 1, \text{ find } f'(-2).$$

$$f(-2) = 4 - 1 = 3.$$

$$f(-2+h) = (-2+h)^2 - 1 = 4 - 4h + h^2 - 1 = 3 - 4h + h^2.$$

$$\text{Subtract } f(-2) = 3: -4h + h^2.$$

$$\text{Divide by } h: -4 + h.$$

$$\text{Let } h \rightarrow 0: -4.$$

$$f'(-2) = -4.$$

Standard

$$7. f(x) = -2x^2, \text{ find } f'(1) \text{ using the } x\text{-form definition.}$$

$$f(1) = -2.$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{-2x^2 - (-2)}{x - 1} = \lim_{x \rightarrow 1} \frac{-2x^2 + 2}{x - 1}$$

$$\text{Factor: } -2x^2 + 2 = -2(x^2 - 1) = -2(x - 1)(x + 1).$$

$$\lim_{x \rightarrow 1} \frac{-2(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} [-2(x + 1)]$$

$$\text{Plug in } x = 1: -2(1 + 1) = -4.$$

$$f'(1) = -4.$$

$$8. f(x) = x^2 - 4x, \text{ find } f'(3) \text{ using the } x\text{-form definition.}$$

$$f(3) = 9 - 12 = -3.$$

$$f'(3) = \lim_{x \rightarrow 3} \frac{(x^2 - 4x) - (-3)}{x - 3} = \lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x - 3}$$

$$\text{Factor the top: numbers multiplying to 3, adding to } -4: -1 \text{ and } -3.$$

$$x^2 - 4x + 3 = (x - 1)(x - 3)$$

$$\lim_{x \rightarrow 3} \frac{(x-1)(x-3)}{x-3} = \lim_{x \rightarrow 3} (x-1)$$

Plug in $x = 3$: $3 - 1 = 2$.

$$f'(3) = 2.$$

9. $f(x) = x^2 + 1$, find $f'(1)$ using the x -form definition, then find the tangent and normal lines at $(1, 2)$.

$$f(1) = 2.$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{(x^2 + 1) - 2}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \rightarrow 1} (x+1) = 2$$

$$f'(1) = 2.$$

Tangent line: $y - 2 = 2(x - 1) \Rightarrow y = 2x$.

Normal line: slope is $-\frac{1}{2}$: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{1}{2} + 2 = -\frac{1}{2}x + \frac{5}{2}$.

Tangent line: $y = 2x$. Normal line: $y = -\frac{1}{2}x + \frac{5}{2}$.

10. $f(x) = 3x^2 - x$, find $f'(2)$ and the tangent line.

$$f(2) = 12 - 2 = 10.$$

$$f(2+h) = 3(2+h)^2 - (2+h) = 3(4+4h+h^2) - 2-h = 12+12h+3h^2-2-h = 10+11h+3h^2.$$

Subtract $f(2) = 10$: $11h + 3h^2$.

Divide by h : $11 + 3h$.

Let $h \rightarrow 0$: 11.

$$f'(2) = 11.$$

Tangent line: $y - 10 = 11(x - 2) \Rightarrow y = 11x - 22 + 10 = 11x - 12$.

11. $f(x) = -x^2 + 2x + 1$, find $f'(0)$ and the tangent line.

$$f(0) = 1.$$

$$f(0+h) = -h^2 + 2h + 1.$$

Subtract $f(0) = 1$: $-h^2 + 2h$.

Divide by h : $-h + 2$.

Let $h \rightarrow 0$: 2.

$$f'(0) = 2.$$

Tangent line: $y - 1 = 2(x - 0) \Rightarrow y = 2x + 1$.

12. $f(x) = 2x^2 - 5x$, find $f'(1)$ and both the tangent and normal lines.

$$f(1) = 2 - 5 = -3.$$

$$f(1+h) = 2(1+h)^2 - 5(1+h) = 2(1+2h+h^2) - 5 - 5h = 2 + 4h + 2h^2 - 5 - 5h = -3 - h + 2h^2.$$

$$\text{Subtract } f(1) = -3: -h + 2h^2.$$

$$\text{Divide by } h: -1 + 2h.$$

$$\text{Let } h \rightarrow 0: -1.$$

$$f'(1) = -1.$$

$$\text{Tangent line: } y - (-3) = -1(x - 1) \Rightarrow y + 3 = -x + 1 \Rightarrow y = -x - 2.$$

$$\text{Normal line: slope is } -\frac{1}{-1} = 1: y + 3 = 1(x - 1) \Rightarrow y = x - 1 - 3 = x - 4.$$

$$\text{Tangent line: } y = -x - 2. \text{ Normal line: } y = x - 4.$$

Challenge

$$13. f(x) = \frac{1}{x+1}, \text{ find } f'(1).$$

$$f(1) = \frac{1}{2}.$$

$$\frac{f(1+h) - f(1)}{h} = \frac{\frac{1}{2+h} - \frac{1}{2}}{h}$$

Combine the top over a common denominator $2(2+h)$:

$$\frac{1}{2+h} - \frac{1}{2} = \frac{2 - (2+h)}{2(2+h)} = \frac{-h}{2(2+h)}$$

So the whole expression is:

$$\frac{-h}{2(2+h)} \cdot \frac{1}{h} = \frac{-1}{2(2+h)}$$

$$\text{Let } h \rightarrow 0: \frac{-1}{2(2)} = -\frac{1}{4}.$$

$$f'(1) = -\frac{1}{4}.$$

$$14. f(x) = \sqrt{x+5}, \text{ find } f'(4).$$

$$f(4) = \sqrt{9} = 3.$$

$$\frac{f(4+h) - f(4)}{h} = \frac{\sqrt{9+h} - 3}{h}$$

Multiply top and bottom by the conjugate $\sqrt{9+h} + 3$:

$$\frac{(9+h)-9}{h(\sqrt{9+h}+3)} = \frac{h}{h(\sqrt{9+h}+3)} = \frac{1}{\sqrt{9+h}+3}$$

Let $h \rightarrow 0$: $\frac{1}{\sqrt{9}+3} = \frac{1}{3+3} = \frac{1}{6}$.

$$f'(4) = \frac{1}{6}.$$

15. $f(x) = x^3 - x$, find $f'(-1)$ using the x -form definition.

$$f(-1) = (-1)^3 - (-1) = -1 + 1 = 0.$$

$$f'(-1) = \lim_{x \rightarrow -1} \frac{(x^3 - x) - 0}{x - (-1)} = \lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1}$$

Factor the top: $x^3 - x = x(x^2 - 1) = x(x - 1)(x + 1)$.

$$\lim_{x \rightarrow -1} \frac{x(x - 1)(x + 1)}{x + 1} = \lim_{x \rightarrow -1} [x(x - 1)]$$

Plug in $x = -1$: $(-1)(-1 - 1) = (-1)(-2) = 2$.

$$f'(-1) = 2.$$

16. Show $f(x) = |x - 2|$ is not differentiable at $x = 2$.

We compute the difference quotient at $a = 2$: since $f(2) = |2 - 2| = 0$,

$$\frac{f(2+h) - f(2)}{h} = \frac{|h| - 0}{h} = \frac{|h|}{h}$$

As $h \rightarrow 0^+$ (small positive h): $|h| = h$, so $\frac{|h|}{h} = \frac{h}{h} = 1$.

As $h \rightarrow 0^-$ (small negative h): $|h| = -h$, so $\frac{|h|}{h} = \frac{-h}{h} = -1$.

Since $1 \neq -1$, the left-hand and right-hand versions of the limit disagree.

Answer: $f'(2)$ does not exist — $f(x) = |x - 2|$ is not differentiable at $x = 2$. This matches the picture: the graph of $|x - 2|$ has a sharp corner (a “V” shape) at $x = 2$, and a sharp corner never has one single well-defined tangent slope.

17. $f(x) = \begin{cases} x^2, & x < 1 \\ 2x - 1, & x \geq 1 \end{cases}$, is this differentiable at $x = 1$?

First, note $f(1) = 2(1) - 1 = 1$ (from the second piece).

Left-hand difference quotient (using the piece valid for $x < 1$, so $h < 0$ meaning $1 + h < 1$): with $f(1+h) = (1+h)^2 = 1 + 2h + h^2$,

$$\frac{f(1+h)-f(1)}{h} = \frac{(1+2h+h^2)-1}{h} = \frac{2h+h^2}{h} = 2+h$$

As $h \rightarrow 0^-$: 2.

Right-hand difference quotient (using the piece valid for $x \geq 1$, so $h > 0$): with $f(1+h) = 2(1+h) - 1 = 1 + 2h$,

$$\frac{f(1+h)-f(1)}{h} = \frac{(1+2h)-1}{h} = \frac{2h}{h} = 2$$

As $h \rightarrow 0^+$: 2.

Both sides give 2.

Answer: $f'(1) = 2$ — f IS differentiable at $x = 1$. The two pieces don't just meet at the same height (which made it continuous, as you found in an earlier unit) — they also arrive with the exact same slope, so they blend together smoothly with no corner at all.

Applied

18. $h(t) = -5t^2 + 30t$, find $h'(1)$.

$$h(1) = -5 + 30 = 25.$$

$$h(1+k) = -5(1+k)^2 + 30(1+k) = -5(1+2k+k^2) + 30 + 30k = -5 - 10k - 5k^2 + 30 + 30k = 25 + 20k - 5k^2.$$

$$\text{Subtract } h(1) = 25: 20k - 5k^2.$$

$$\text{Divide by } k: 20 - 5k.$$

$$\text{Let } k \rightarrow 0: 20.$$

$h'(1) = 20$. This means at exactly $t = 1$ second, the ball is rising at an instantaneous speed of 20 meters per second.

19. $C(x) = 0.5x^2 + 10x$, find $C'(4)$.

$$C(4) = 0.5(16) + 40 = 8 + 40 = 48.$$

$$C(4+k) = 0.5(4+k)^2 + 10(4+k) = 0.5(16+8k+k^2) + 40 + 10k = 8 + 4k + 0.5k^2 + 40 + 10k = 48 + 14k + 0.5k^2.$$

$$\text{Subtract } C(4) = 48: 14k + 0.5k^2.$$

$$\text{Divide by } k: 14 + 0.5k.$$

$$\text{Let } k \rightarrow 0: 14.$$

$C'(4) = 14$. This means when the company is producing around 4 units, each additional unit costs about \$14 more to produce — this is called the marginal cost at that production level.

20. $P(t) = t^2 + 50t$, find $P'(3)$.

$$P(3) = 9 + 150 = 159.$$

$$P(3+k) = (3+k)^2 + 50(3+k) = 9 + 6k + k^2 + 150 + 50k = 159 + 56k + k^2.$$

$$\text{Subtract } P(3) = 159: 56k + k^2.$$

$$\text{Divide by } k: 56 + k.$$

$$\text{Let } k \rightarrow 0: 56.$$

$P'(3) = 56$. This means at year 3, the population is growing at an instantaneous rate of 56 people per year.

$$21. \ s(t) = 4t^2 - 2t, \text{ find } s'(2).$$

$$s(2) = 16 - 4 = 12.$$

$$s(2+k) = 4(2+k)^2 - 2(2+k) = 4(4 + 4k + k^2) - 4 - 2k = 16 + 16k + 4k^2 - 4 - 2k = 12 + 14k + 4k^2.$$

$$\text{Subtract } s(2) = 12: 14k + 4k^2.$$

$$\text{Divide by } k: 14 + 4k.$$

$$\text{Let } k \rightarrow 0: 14.$$

$s'(2) = 14$. This means at exactly $t = 2$ seconds, the object's instantaneous speed is 14 meters per second.